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Endpoint Inventory

The API contract, written before the code. This is [[delivery-plan]] Phase 3, and it is where the two workstreams meet and nowhere else. The client is built after the backend, so this file is read by the backend first and by the UI developer later; it must be correct for both.

Status: design, nothing implemented. Once code exists, `docs/api/openapi.json` generated from that code becomes the machine-readable contract and this file becomes the human overview. They must not diverge; a change lands in both in the same pull request.

Rules this document obeys: [[rest-api-design]] · [[openapi-documentation]] · [[signalr-realtime]] · [[authentication-authorization]] · [[owasp-api-top10-checklist]]

Conventions

Aspect	Value
Base path	`/api/v1`
Format	JSON, `camelCase`, UTF-8, UTC ISO-8601 instants
Errors	RFC 9457 `application/problem+json`, one shape, always with `traceId`
Collections	`{ "items": [...], "page": 1, "pageSize": 20, "total": 137 }`, `pageSize` capped at 100
Ids in JSON	strings, always
Auth	`Authorization: Bearer <access token>`
Mutation headers	`X-Client-Platform`, `X-Client-Device`, `X-Client-App-Version` — **required and enforced** since SPEC-008: middleware refuses any `POST`, `PUT`, `PATCH` or `DELETE` that omits one, or whose value exceeds 64 characters, with a bodiless `400`. Reads and the health probes are unaffected. Each accepted request leaves one audit record carrying these three values (SDD §13.4)
Optimistic concurrency	`ETag` on the resource, `If-Match` on the mutation, mismatch means `412`

Authorisation notation

Two gates, both evaluated server-side on every call ([[sdd]] §13.2):

Symbol	Gate 1 — identity	Gate 2 — visibility and licence
`anon`	none	none
`auth`	any authenticated user	none
`self`	the caller owns the resource	none
`owner`	the caller owns the aggregate	none
`member`	the caller is an attendee or collaborator	visibility resolved first
`licence`	the caller passes the aggregate's `LicenceMode` check	includes presence proof where required

A resource the caller may not see returns `404`, never `403`. `403` would confirm it exists.

Status codes used

`200` ok · `201` created (+ `Location`) · `202` accepted (generic, non-enumerating) · `204` no content · `400` malformed or missing client headers · `401` unauthenticated · `403` authenticated but not permitted · `404` absent **or hidden from this caller** · `409` conflict · `412` precondition failed (stale `If-Match`) · `422` semantically invalid · `429` rate limited

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1. Identity — `/api/v1/auth`

Subject V.1, V.5. Rules: [[authentication-authorization]], [[social-oauth]].

Method	Path	Body	Success	Failures	Auth
POST	`/auth/registrations`	`email`, `password`, `displayName`	`202`	`400`, `422`, `429`	`anon`
POST	`/auth/email-verifications`	`token`	`204`	`400`, `410` expired, `429`	`anon`
POST	`/auth/verification-emails`	`email`	`202`	`429`	`anon`
POST	`/auth/sessions`	`email`, `password`	`200` token pair	`401` generic, `429`	`anon`
POST	`/auth/sessions/social`	`provider`, `idToken`	`200` token pair	`401`, `422`, `429`	`anon`
POST	`/auth/tokens`	`refreshToken`	`200` rotated pair	`401`, `429`	`anon`
DELETE	`/auth/sessions/current`	—	`204`	`401`	`auth`
GET	`/auth/sessions`	—	`200` list	`401`	`self`
DELETE	`/auth/sessions/{sessionId}`	—	`204`	`401`, `404`	`self`
POST	`/auth/password-reset-requests`	`email`	`202`	`429`	`anon`
POST	`/auth/password-resets`	`token`, `newPassword`	`204`	`400`, `410`, `422`, `429`	`anon`
GET	`/auth/social-links`	—	`200` list	`401`	`self`
POST	`/auth/social-links`	`provider`, `idToken`	`201`	`409` already linked, `422`	`auth`
DELETE	`/auth/social-links/{provider}`	—	`204`	`409` last sign-in method	`self`

OpenID Provider surface — S8 The API becomes an OpenID Provider at S8 – [[ADR-0031-musicroom-as-openid-provider]]. Paths and parameter names below follow OpenID Connect Core 1.0 and are **not** ours to rename.

Method	Path	Request	Success	Errors	Auth
GET	`/.well-known/openid-configuration`	—	200 discovery document	—	`anon`
GET	`/.well-known/jwks.json`	—	200 public key set	—	`anon`
GET	`/oauth2/authorize`	`response_type=code`, `client_id`, `redirect_uri`, `scope`, `state`, `nonce`, `code_challenge`, `code_challenge_method=S256`	302 to `redirect_uri` with `code`	302 with `error`	`anon`
POST	`/oauth2/token`	`grant_type=authorization_code` + `code` + `code_verifier`, or `grant_type=refresh_token` + `refresh_token`	200 access token + ID token + rotated refresh token	400 `invalid_grant`, 401, 429	`anon`
GET	`/oauth2/userinfo`	—	200 claims the granted scopes allow	401	`auth`
POST	`/oauth2/revoke`	`token`	200	401	`anon`

`code_challenge_method=S256` is required; `plain` is refused. There is no `grant_type=password`: OAuth 2.1 removes the resource-owner password-credentials grant and RFC 9700 deprecates it.

These replace rather than join `POST /auth/sessions`, `POST /auth/sessions/social` and `POST /auth/tokens`. The removal is part of S8's spec, and no old endpoint is deleted before its replacement passes.

Design notes

- **`202` on registration, verification resend and password reset is deliberate.** A `409 email already exists` is an account-enumeration oracle. The response is identical whether or not the address is known; the mailbox tells the real story. Cost: a typo'd address gives no immediate feedback. Accepted, and stated at defence.
- `/auth/tokens` performs refresh **rotation**. Presenting an already-rotated refresh token revokes the whole family and returns `401` — this is the session-theft detector (T-03).
- `/auth/password-resets` revokes every refresh token for that account on success.
- **A session is a refresh-token family.** `GET /auth/sessions` returns one entry per live family with the `platform`, `device` and `appVersion` that started it, when it started and when it stops being renewable — never a token or a digest. `DELETE /auth/sessions/current` ends the family named by the access token's `sid` claim, so signing out of one device leaves the others signed in. `DELETE /auth/sessions/{sessionId}` ends a named one, and answers `404` for an id that is unknown, already ended or another account's.
- **An access token outlives a logout** by up to 15 minutes and 30 seconds, which is its remaining lifetime plus the accepted clock skew. Revoking the family stops renewal, not a token already signed — recorded as T-28.

- `/auth/sessions/social` both registers and signs in. Identity comes from the validated **ID token**, never an access token ([[ADR-0008-oidc-social-login]]).
- `DELETE /auth/social-links/{provider}` refuses with `409` when it would leave the account with no usable sign-in method.
- Every row here is rate limited per account **and** per IP.

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2. Profile and social graph — `/api/v1/me`, `/api/v1/users`

Subject V.1. The friends graph exists because the `Friends` visibility tier requires it.

Method	Path	Body	Success	Failures	Auth
GET	`/me`	—	`200` full own profile	`401`	`self`
PATCH	`/me/profile`	any subset of profile fields, each with its `visibility`	`200`	`422`	`self`
PUT	`/me/music-preferences`	`genres[]`, `artists[]`	`200`	`422`	`self`
GET	`/users`	query: `q`, `page`, `pageSize`	`200` paged, public fields only	`401`	`auth`
GET	`/users/{userId}`	—	`200` **filtered per field**	`404`	`auth`
GET	`/me/friends`	paged	`200`	`401`	`self`
DELETE	`/me/friends/{userId}`	—	`204`	`404`	`self`
GET	`/me/friend-requests`	query: `direction=incoming|outgoing`	`200` paged	`401`	`self`
POST	`/me/friend-requests`	`targetUserId`	`201`	`409` duplicate or already friends, `404`	`auth`
PATCH	`/me/friend-requests/{requestId}`	`status`: `accepted | declined`	`200`	`404`, `409`	`self` (recipient)
DELETE	`/me/friend-requests/{requestId}`	—	`204`	`404`	`self` (sender)

Design notes

- `GET /users/{userId}` projects **per field** according to the requester's relationship (self / friend / stranger) against each field's `ProfileFieldVisibility`. The full row never leaves the server for the client to filter — OWASP API3.
- `PATCH /me/profile` sets both value and visibility in one call, so a new field cannot exist without a tier. A missing tier is `422`, never a silent default to public.
- `GET /users` search returns public fields only, paginated, rate limited. Unbounded user enumeration is OWASP API4.

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3. Catalog — `/api/v1/tracks`

Subject III. Rules: [[youtube-integration]].

Method	Path	Body	Success	Failures	Auth
GET	`/tracks`	query: `q`, `page`, `pageSize`	`200` paged `TrackRef`	`429`, `503` quota	`auth`
GET	`/tracks/{youtubeVideoId}`		`200` `TrackRef`	`404`, `503`	`auth`

Design notes

- `GET /tracks` is the **only** path that can reach `search.list`, the expensive YouTube call. It is debounced client-side, cached server-side, and rate limited. `GET /tracks/{id}` prefers the cache, then `videos.list`.
- The YouTube API key is server-side only. The client never calls YouTube for metadata.
- Quota exhaustion surfaces as `503` with a clear problem type, not a `500`.
- A `TrackRef` is `{ youtubeVideoId, title, channel, durationSeconds, thumbnailUrl }`. No media.

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4. Track Vote — `/api/v1/events`

Subject V.2.1. The contended write path. Rules: [[realtime-architecture]], [[ADR-0014-location-time-licence]].

Method	Path	Body	Success	Failures	Auth
POST	`/events`	`name`, `visibility`, `licence`, `startsAtUtc`, `endsAtUtc`	`201`	`422`	`auth`
GET	`/events`	query: `scope=public|mine|invited`, paging	`200` paged	`401`	`auth`
GET	`/events/{eventId}`		`200`	`404` if hidden	`member`
PATCH	`/events/{eventId}`	metadata subset	`200`	`404`, `412`	`owner`
DELETE	`/events/{eventId}`		`204` soft delete	`404`	`owner`
PUT	`/events/{eventId}/visibility`	`visibility`, `licence`, `mode`, `presenceProof`, `geofence`, `window`	`200`	`422`	`owner`
POST	`/events/{eventId}/join-codes`	`ttlSeconds`	`201` short-lived code	`404`, `429`	`owner`
GET	`/events/{eventId}/invitations`	paging	`200`	`404`	`owner`
POST	`/events/{eventId}/invitations`	`userId`	`201`	`404`, `409`	`owner`
DELETE	`/events/{eventId}/invitations/{userId}`		`204`	`404`	`owner`
POST	`/events/{eventId}/attendances`		`201`	`404`, `409`	`auth` if public, `member` if private
DELETE	`/events/{eventId}/attendances/current`		`204`	`404`	`self`
GET	`/events/{eventId}/queue`	paging, vote-ordered	`200`	`404`	`member`
POST	`/events/{eventId}/tracks`	`youtubeVideoId`	`201` suggestion	`404`, `409` duplicate, `422`	`licence`

Method	Path	Body	Success	Failures	Auth
DELETE	`/events/{eventId}/tracks/{suggestionId}`		204	404	`owner` or suggester
PUT	`/events/{eventId}/tracks/{suggestionId}/vote`	optional `presence`	201 / 200	404, 409, 422, 429	`licence`
DELETE	`/events/{eventId}/tracks/{suggestionId}/vote`		204	404	`self`
PATCH	`/events/{eventId}/queue/now-playing`	`{ suggestionId }`	200	404, 409	`owner` or delegate

Design notes

- **Voting is `PUT`, not `POST`.** A vote is an idempotent fact: "this user votes for this track". A double submit from a flaky mobile network is then harmless — `201` the first time, `200` after. `POST` would create a second row and make the race worse.
- The vote's uniqueness is enforced by `ux_votes_event_id_track_id_user_id`, and the duplicate is detected **by the database**, never by a read-then-write check ([[ef-core-persistence]]).
- The vote count is recomputed from the source rows inside the transaction. Nothing increments a counter held in memory.
- `presence` carries `{ coordinates?, eventCode? }` and is required only when the licence is `LocationAndTimeRestricted`. The server checks it against the event's geofence and window, using **server time**. The coordinate is used and discarded — it is never written to the vote row (threat T-21) and never logged.
- `POST /events/{eventId}/event-codes` is owner-only and rate limited; codes are short-lived and single-use per user.
- The broadcast on `EventHub` happens **after** commit, never before.
- **Attendance is not a gate on a public event.** Subject V.2.1 says that when an event is public, *all* users can find it and vote. `POST /attendances` records presence for the roster; it is never required before `PUT .../vote` on a public event. On a private event, the invitation is the gate, not the attendance.

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5. Playlist Editor — `/api/v1/playlists`

Subject V.2.3. The second contended write path.

Method	Path	Body	Success	Failures	Auth
POST	`/playlists`	`name`, `visibility`, `licence`	201	422	`auth`
GET	`/playlists`	query: `scope=public|mine|shared`, paging	200 paged	401	`auth`
GET	`/playlists/{playlistId}`		200 + `ETag`	404 if hidden	`member`
PATCH	`/playlists/{playlistId}`	metadata subset	200	404, 412	`licence`
DELETE	`/playlists/{playlistId}`		204 soft delete	404	`owner`
PUT	`/playlists/{playlistId}/licence`	`licence`, `mode`	200	422	`owner`
GET	`/playlists/{playlistId}/collaborators`		200	404	`member`
POST	`/playlists/{playlistId}/collaborators`	`userId`, `canEdit`	201	404, 409	`owner`
DELETE	`/playlists/{playlistId}/collaborators/{userId}`		204	404	`owner`
GET	`/playlists/{playlistId}/tracks`	paging	200 + `ETag`	404	`member`

Method	Path	Body	Success	Failures	Auth
POST	`/playlists/{playlistId}/tracks`	`youtubeVideoId`, `afterTrackId?`	`201`	`404`, `422`	`licence`
PATCH	`/playlists/{playlistId}/tracks/{playlistTrackId}`	`If-Match`: `ETag`; `afterTrackId` or `beforeTrackId`	`200`	`404`, `409`, `412`	`licence`
DELETE	`/playlists/{playlistId}/tracks/{playlistTrackId}`	—	`204`	`404`	`licence`

Design notes

- **Reorder is a `PATCH` with `If-Match`.** The client sends the `ETag` it last read; a stale one gets `412` and the client reloads. Without this, two simultaneous moves silently overwrite each other — the exact failure subject V.2.3 grades.
- Position is a sparse ordering key computed **between the neighbours named in the request**, so inserting between two tracks does not rewrite the list.
- `afterTrackId: null` means "move to the head". Sending both `afterTrackId` and `beforeTrackId` is `422`.
- Conflicts that survive a reload-and-reapply return `409` rather than guessing the user's intent.
- `PlaylistHub` broadcasts `TrackMoved` with the new version after commit.

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6. Control Delegation — `/api/v1/devices`

Subject V.2.2. Licence is **per device**, bound to the account.

Method	Path	Body	Success	Failures	Auth
POST	`/devices`	`platform`, `model`, `appVersion`	`201`	`409` already registered, `422`	`auth`
GET	`/devices`	—	`200` own devices	`401`	`self`
DELETE	`/devices/{deviceId}`		`204`	`404`	`self`
GET	`/devices/{deviceId}/control`		`200` + `ETag` current controller	`404`	`self` or controller
PUT	`/devices/{deviceId}/control`	`granteeUserId`	`200`	`404`, `409`, `412`	`owner`
DELETE	`/devices/{deviceId}/control`		`204` released	`404`	`owner` or controller
GET	`/me/control-grants`	—	`200` grants held by me	`401`	`self`

Design notes

- Control is a **single row with a state machine and a concurrency token** ([[sdd]] §11.2). Two simultaneous claims: one wins, the other gets `412`. The loser must lose **visibly** — a silent overwrite is the bug.
- `granteeUserId` must be a friend of the owner. Otherwise `422`.
- `DELETE .../control` is allowed to both the owner (revoke) and the current controller (release).
- `POST /devices` derives its values from the client context headers, which are mandatory anyway.

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7. Playback — `/api/v1/playback-sessions`

Subject V.2.1 (a highly-voted track plays earlier) and V.2.2 (control is per device). Model and rationale: [[ADR-0017-playback-session]].

Method	Path	Body	Success	Failures	Auth
POST	`/playback-sessions`	`deviceId`, `sourceType`, `sourceId`	`201` + `ETag`	`404`, `409` device already has a session, `422`	`owner` of the device
GET	`/playback-sessions/{sessionId}`		`200` + `ETag`	`404`	`member` of the source
GET	`/devices/{deviceId}/playback-session`		`200` + `ETag`	`404`	`self` or controller
POST	`/playback-sessions/{sessionId}/advance`	`expectedNowPlayingId`	`200` now-playing	`404`, `409` queue empty, `412` stale	controller
PATCH	`/playback-sessions/{sessionId}`	`nowPlayingId` (explicit jump)	`200`	`404`, `412`	controller
DELETE	`/playback-sessions/{sessionId}`		`204` stop playback	`404`	`owner` of the device

Design notes

- **`expectedNowPlayingId` makes advance idempotent.** A duplicate call from a flaky mobile network finds its expectation stale and gets the current state back — it never skips a second track. Without it, a retry silently loses a song.
- The server picks the next item: for an `Event`, the highest vote count with ties broken by earliest suggestion; for a `Playlist`, the next ordering key. **The client reports that the track ended; it never chooses what plays next.**
- `controllerUserId` on the session is what `PUT /devices/{id}/control` changes. That is the concrete meaning of "delegating music control".
- `NowPlayingChanged` broadcasts on `EventHub` or `PlaylistHub` **after** commit.
- A device holds at most one active session.

8. Audit and operations

Method	Path	Success	Auth	Note
GET	`/me/activity`	`200` paged own audit records	`self`	optional; the **write** side is the graded requirement (V.6), the read side is convenience
GET	`/health`	`200`	`anon`	liveness, minimal body
GET	`/health/ready`	`200` / `503`	`anon`	readiness: database and provider reachability
GET	`/openapi/v1.json`	`200`	`anon`	the generated contract; interactive UI only in `Development`

Audit records are **written** by a pipeline behaviour on every command — that is subject V.6, and it is not optional. See [[logging-observability]].

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9. Realtime — client-to-server hub methods

Server-to-client events are in [[sdd]] §9.2 and [[signalr-realtime]]. Hub methods are API endpoints with a different transport, and the same authorisation applies.

Hub	Method	Effect	Auth
`EventHub`	`JoinEvent(eventId)`	join group `event:{eventId}`	visibility + licence, **re-checked on reconnect**
`EventHub`	`LeaveEvent(eventId)`	leave the group	`auth`
`PlaylistHub`	`JoinPlaylist(playlistId)`	join group `playlist:{playlistId}`	visibility + licence, re-checked on reconnect
`PlaylistHub`	`LeavePlaylist(playlistId)`	leave the group	`auth`
`ControlHub`	—	auto-joins `user:{userId}` on connect	`auth`

Design notes

- **Mutations do not happen over the hub.** Every state change goes through the REST endpoints above, so one authorisation pipeline and one validation pipeline cover both transports. The hub carries joins, leaves and server-pushed facts.
- A client asking to join `playlist:42` proves nothing. Group membership is authorised server-side, every time.
- Authentication uses the same JWT, via the documented SignalR `access_token` query parameter on the WebSocket handshake.
- **Client context is captured at the handshake**, not per invocation — a hub call carries no HTTP headers. A handshake without `platform`, `device` and `appVersion` is rejected, mirroring the `400` a REST mutation gets for the same omission.
- **Join and leave are audited** by a hub filter writing the same record as the REST behaviour ([[ADR-0019-hub-action-auditing]]). Connect and disconnect are not — mobile reconnect noise would bury deliberate actions.

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10. What is deliberately absent

Not present	Why
Any endpoint returning another user's private data	subject V.6; there is no "admin view" in this project
Bulk or unpaginated list endpoints	OWASP API4; every list is capped
A `PUT /users/{id}` style full replace	invites mass assignment; profile updates are `PATCH` with explicit fields
Media streaming or proxying	playback is client-side YouTube; we store only `TrackRef`
Direct database or Supabase access from a client	subject V.3 makes the backend the sole guardian of truth
Mutation over SignalR	keeps one authorisation and validation path

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11. Open items

#	Item	Resolved by
E1	Concrete request and response schemas per endpoint	the OpenAPI document, generated from code
E2	Exact `PlaybackSession` and control state names	the S7 slice ([[sdd]] §11.2)
E3	Whether `GET /me/activity` ships	drop it first if S9 is tight; the write side is what is graded
E4	Pagination cursor vs offset for the event queue	offset assumed; revisit if the queue grows large

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Related

[[sdd]] §9 · [[delivery-plan]] · [[rest-api-design]] · [[signalr-realtime]] · [[owasp-api-top10-checklist]] · [[musicroom-backend]] · [[musicroom-mobile]]